Statistical K-Means with PCA — DJF Morocco

1 Method

I implemented a purely statistical K-Means clustering to identify four DJF weather regimes from ERA5 Z500 anomalies over the North Atlantic (1979–2020). No precipitation is used during clustering.

Before clustering, I applied PCA to compress each daily Z500 map from 825 grid points to a small number of independent components (EOFs). PCA is fitted on the 30 training winters only and then applied to both sets. K-Means is then fitted in the EOF space on training days only, with k=4.

The four clusters are ordered wettest to driest using training precipitation only, so that Regime 1 always corresponds to the pattern most associated with extreme rainfall over Morocco. The workflow is shown in Figure 1.

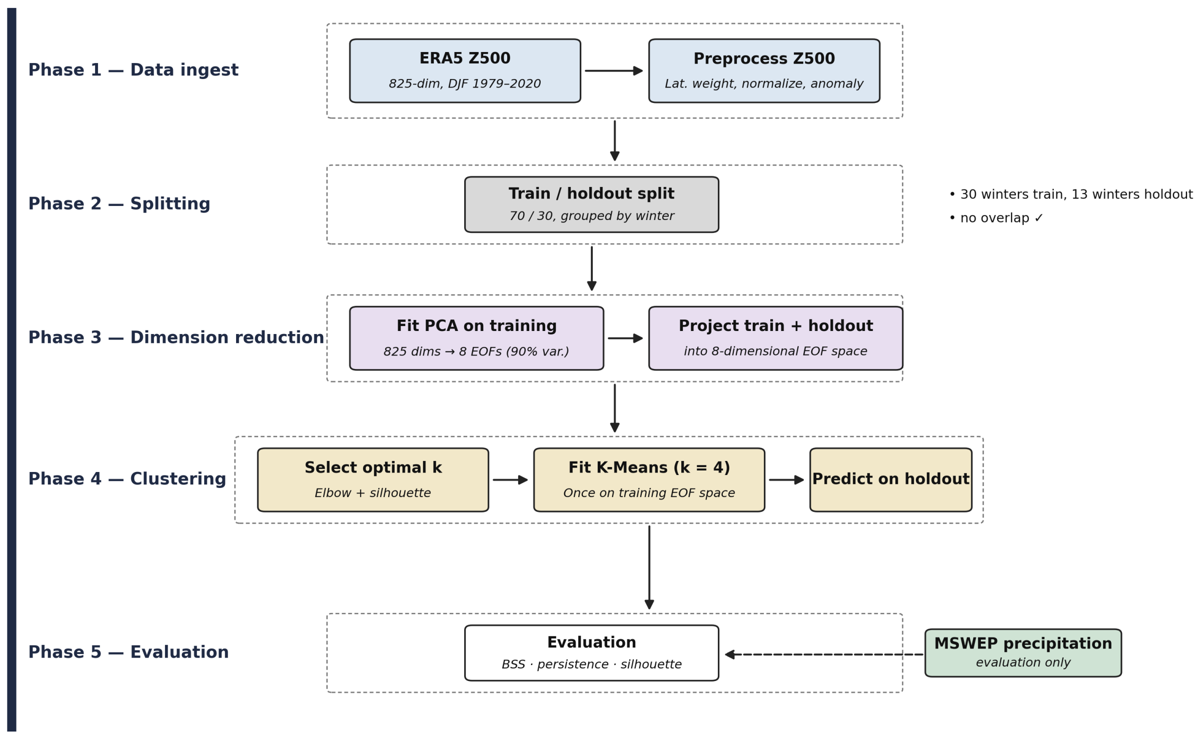

Figure 1: Workflow of the statistical K-Means method.

2 Results

2.1 Regime Cluster Centers

The Z500 composite maps of the four regimes obtained on the training set are shown in Figure 2.

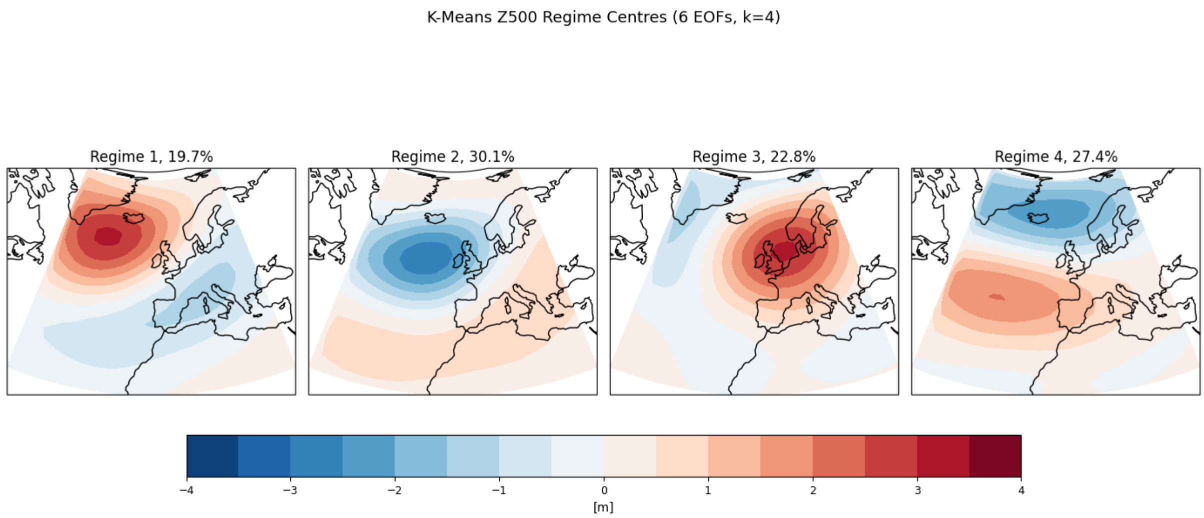

Figure 2: Z500 composite maps of the four regimes (training winters).

2.2 Odds Ratio Maps

Figure 3 shows the Q95 precipitation odds ratio per regime computed on the holdout set.

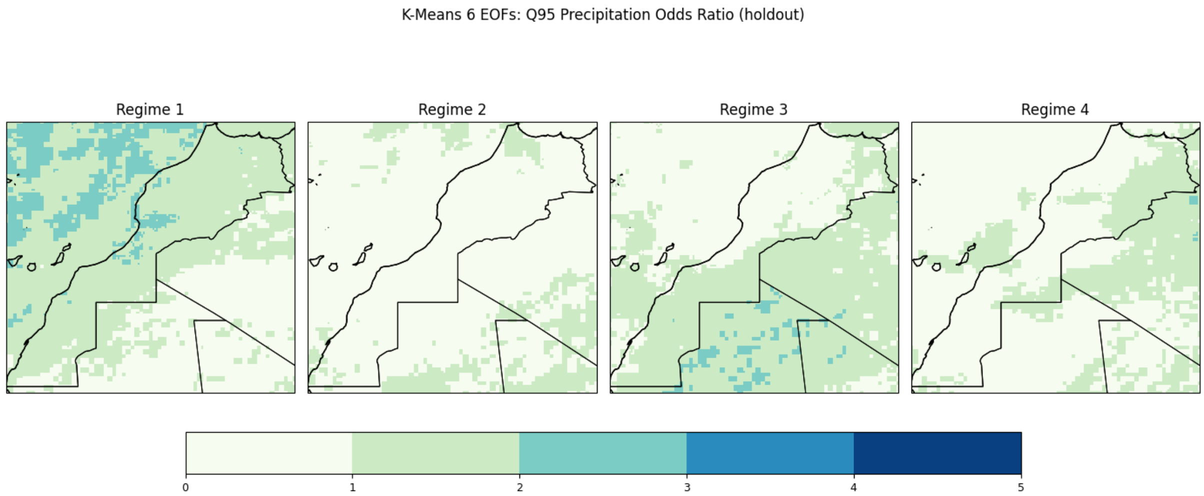

Figure 3: Q95 precipitation odds ratio by regime (holdout winters).

2.3 Evaluation

Table 1 summarises the key evaluation metrics.

Table 1: Evaluation results (6 EOFs).

Metric	**Value**
Silhouette score (training)	0.1801
Avg regime persistence (holdout)	6.9 days
BSS 95th percentile (holdout)	0.0273
BSS Cluster (holdout)	0.0470

The positive BSS scores confirm that the regimes carry predictive information about Morocco precipitation on unseen winters.

3 Number of EOFs: 6 vs 8

The results presented in this report are obtained using 6 EOFs. At first, I considered using 8 EOFs because they explain around 90.2% of the total variance and this choice is often used in the literature. However, from the explained variance curve, the elbow appears around EOF 5–6, and 6 EOFs already explain about 84.2% of the variance.

After EOF 6, the additional EOFs contribute less variance, which may indicate that they represent smaller-scale variability or noise rather than the dominant circulation patterns.

For this reason, I decided to use 6 EOFs for the clustering results shown in this report. However, I would like your opinion on whether it is better to keep 6 EOFs as a simpler representation of the data, or use 8 EOFs to include more variability before applying K-Means clustering.

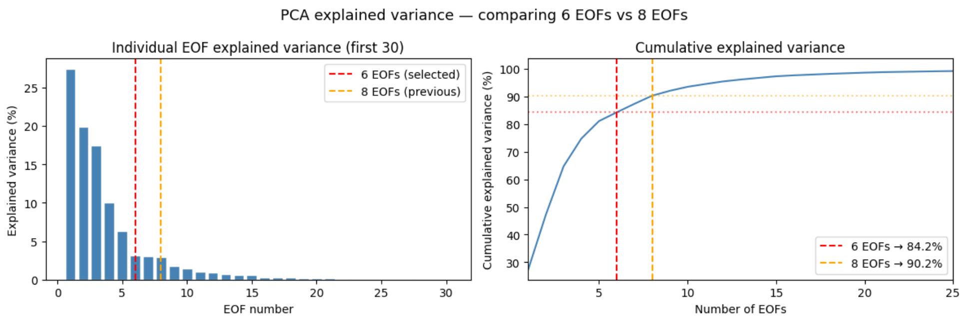

Figure 4: Cumulative explained variance. The red marker (6 EOFs, 84.2%) and the orange marker (8 EOFs, 90.2%) are both shown.

4 Literature Review: Mediterranean EPEs Study

In parallel with this work, I have been reading the paper *"Extreme precipitation events in the Mediterranean: Spatiotemporal characteristics and connection to large-scale atmospheric flow patterns"*.

The main goal of this paper is to understand and quantify the connection between Extreme Precipitation Events (EPEs) in the Mediterranean and large-scale atmospheric weather patterns.

To achieve this goal, the paper outlines four specific research questions:

- Do EPEs over the Mediterranean dominantly occur in specific seasons?
- To which degree are EPEs spatially and temporally connected?
- Do results based on ERA5 precipitation agree with those of previous studies using other precipitation datasets?
- How useful is weather clustering over the whole Mediterranean domain for conditioning localized EPEs?